

The Analysis of MMA fights and Fighters

MASON PARK*, Kennesaw State University, USA

ACM Reference Format:

Mason Park. 2026. The Analysis of MMA fights and Fighters. 1, 1 (April 2026), 7 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Abstract

2 Changes from Deliverable M2

Since M2 the changes that were made were minor such as adding EDA techniques in the notebook and in this very document. Another change that was conducted was removing large swathes of data points from the given data sets to clean up the data properly. As a result the heatmap, bar graph, and even the elbow graph were altered as takedown attempts and general significant strikes were equalized. A silhouette score was calculated to validate the k score and the new k score is 3 instead.

Another minor change was narrowing the scope of my discovery question for utilizing the association rule. Asking about the relationship in lower fighter reach, weight class, and stance leading to greater or lesser strike accuracy and bout outcome.

3 EDA Discoveries

The data reveals a clear change over time in the sport, MMA has moved from an era defined by massive disparities in ground control to a more balanced, striking-centric era. While fighters in the early 2000s relied on controlling their opponents for minutes at a time, modern fighters separate themselves through impact and damage (knockdowns), as the defensive wrestling of the average fighter has significantly improved, shrinking the control time gap.

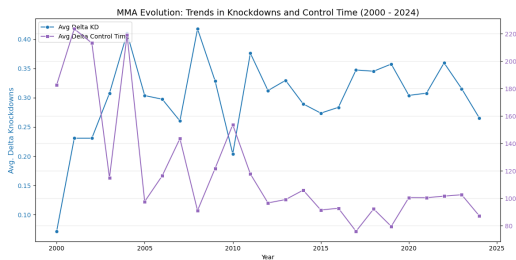


Fig. 1. A short graph showcasing the temporal trend of knockdowns to control times

Author's Contact Information: Mason Park, spark87@students.kennesaw.edu, spark87@students.kennesaw.edu, Kennesaw State University, Marietta, Georgia, USA.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM XXXX-XXXX/2026/4-ART

<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

As well as this elbow graph using the silhouette method for calculating k means the most optimal k mean is 3 or 9 but 3 was chosen for scalability.

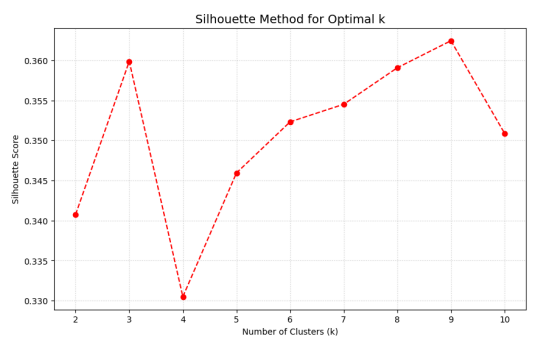


Fig. 2. Silhouette Method showcasing optimal k values

Another case of EDA made with the heatmap pictured below From what is seen on the heat map visualization In answering my question Southpaw and Orthodox have similar distributions in reach teetering between shorter or equal whereas switch differs by having a high density in equal length (within 2 inches difference)

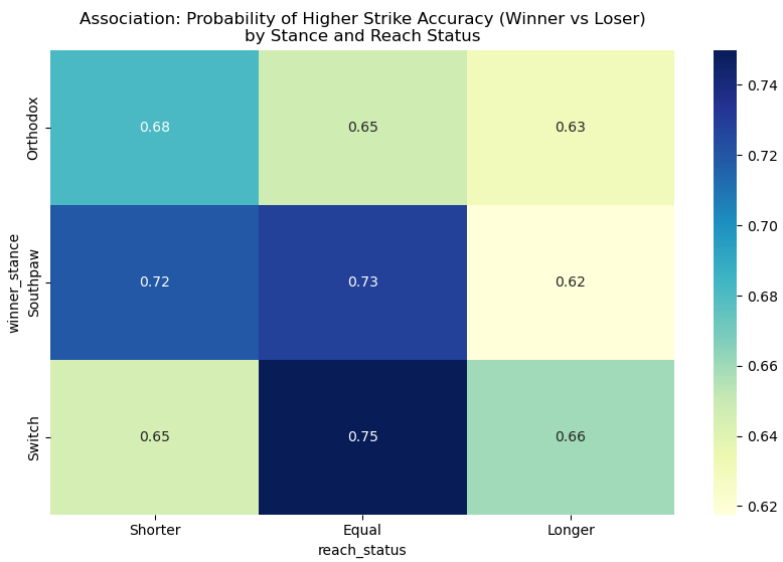


Fig. 3. A heatmap that points out accuracy for strike and reach status.

Providing more visualizations of how these fight outcomes go with a bar graph pictured below questioning the outcomes in bouts by weight class and stance and reach length. From what is seen, While the majority of the datasets are in orthodox stance by weight classes it showcases that each weight class has equal distributions of twin peak graphs almost equalizing between short reach and long reach. While switch stance winners are often longer in certain weight brackets showing they may potentially rely on having a reach advantage.

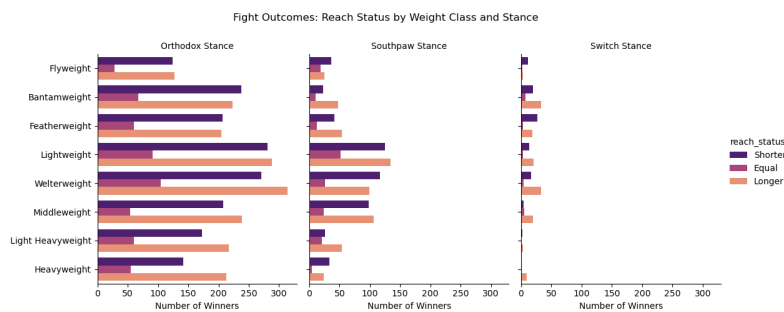


Fig. 4. A bar graph that separates the 3 stances by weight class and measures which reaches won in a bout

4 Analytical Choices

For my first discovery question, asking if weight classes affect fighting style, I used the clustering technique. I used this technique because striking and grappling as viewed in the temporal trend above are independent of one another and are not linear and would be easier to parse when broken down into clusters by weight classes. Showcasing potentially which weight class favors which style and how these styles shift as the weight increases.

For the second discovery question, asking if a fighter's weight class, having a shorter reach, and stance determines strike accuracy and bout outcomes. Targeting important traits that individually do not provide strong correlation for outcomes but together can piece together a method for the desired outcomes.

5 Analysis of Data Results

In answering the first discovery question utilizing clustering for how weight classes determine fighting styles. Some insights were made. At first using Striking Accuracy vs Take down attempts as a metric to read from.

After using an elbow method and the silhouette method for my data to see what the k value should be, it was determined that k = 3 is the most ideal number to use as seen below as k = 3 is where the data points begin to level off.

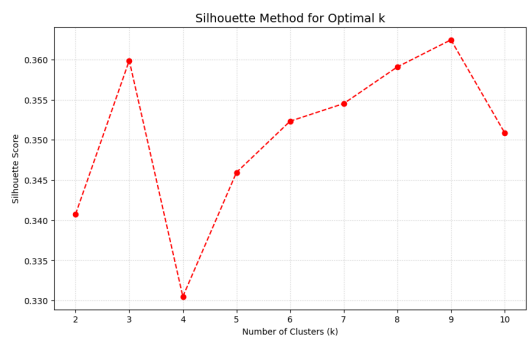


Fig. 6. Silhouette Method showcasing which values are best suited for the k value

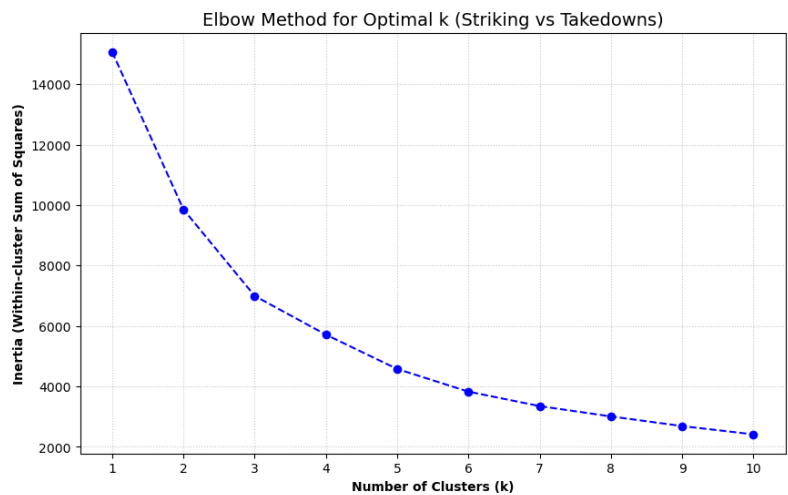


Fig. 5. Elbow Graph from code.

After finding the k value needed for clusters I used two separate graphs to visualize the data better to showcase how weight classes determine fighting style. In a more simplistic visualization using bar charts that separate by weight classes as shown below. With the data being strike accuracy vs take down attempts.

In the bar chart it showcases that almost every style with the exception of flyweight title bouts have higher takedown averages with some lighter weight classes having a greater difference in strike accuracy to takedown attempts while as the weight classes get heavier it begins to equal out a bit while still leaning in favor of takedown attempts.

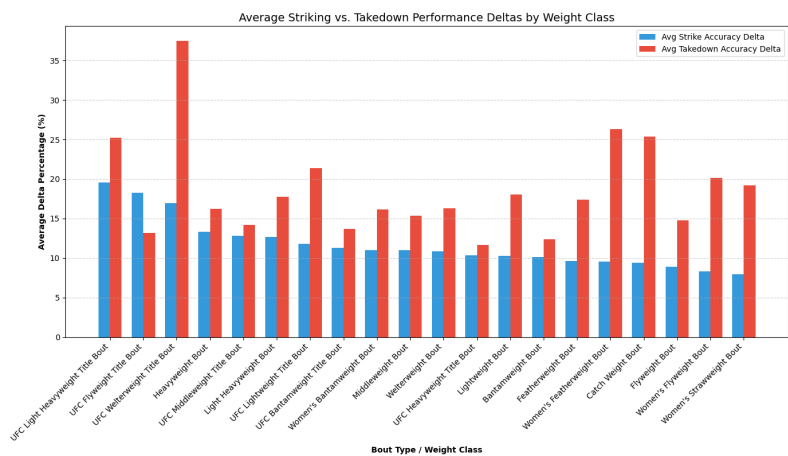


Fig. 7. Bar graph of fight events of strike accuracy vs take down attempts separated by weight class

Another graph I constructed in my code is creating a heat map to showcase the population densities of the clusters. Separating into 3 clusters, 0 showcases grappling heavy fights, 1 showcases balanced a mix between both styles, and the last cluster 2 showcases a fight more heavy into striking. As shown below the heatmap, there seems to be a balance between all styles in lower weight classes however, heavy weight classes seem to be more predominantly striking heavy.

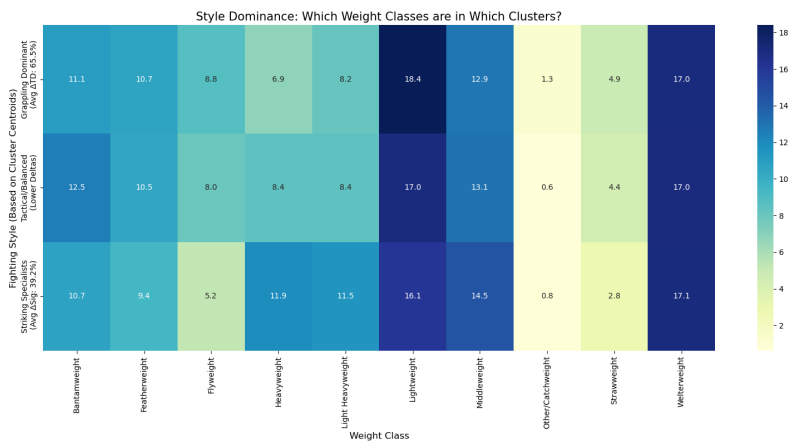


Fig. 8. A heat map showcasing the heatmap of favored styles by weight classes

In answering a second discovery question looking into a relationship of how weight classes, shorter reach, and stance affect strike accuracy and fight outcome. The first heatmap made was determining the outcomes of fights between fighters of these weight classes with shorter reach and sorted by which stance was used. In this heatmap specifically examining weight class, shorter reach, and stance. For Orthodox having a shorter reach doesn't affect the outcome of a bout but on the other hand, southpaw stance for lightweight, welterweight, and middle weight have higher rates of winning if they have a shorter reach. Showcasing that these fighters are more likely to have an advantage if they are using southpaw with a shorter reach.

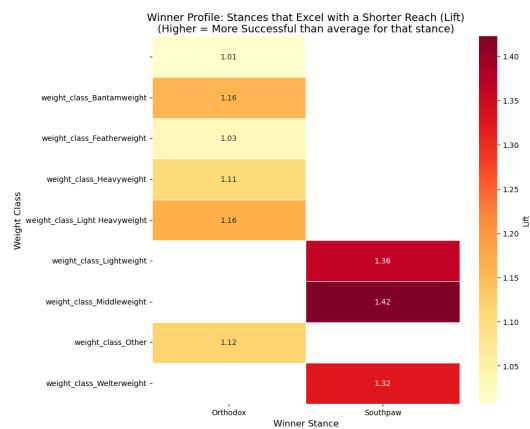


Fig. 9. A heatmap that calculates the outcome of fights and what fighter with a shorter reach used which stance

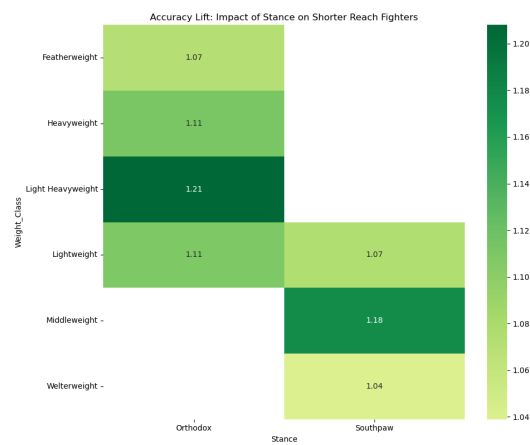


Fig. 10. A heatmap that calculates the strike accuracy of fights and what fighter with a shorter reach used which stance

Another heatmap that was made was the same values but measuring strike accuracy using the number of significant strikes made. Showcasing interestingly different results. In the heatmap below while the weight classes that significantly lean in favor of shorter reach to give high accuracy is light heavyweight for orthodox and middleweight for southpaw showcasing that these weight classes perform better with their respective stance of Orthodox Light Heavyweight and Southpaw middleweight fighters. Showcasing that light heavyweight fighters with the orthodox stance have more accurate strikes much like middleweight fighters with the southpaw stance. Potentially showcasing that other fighters may rely on either having a reach advantage or rely on a volume of strikes rather than surgical strikes.

6 Final Timetable

After the complete implementation I will work on a power point for the presentation and a poster design for another potential presentation. After working on presentation prep if time permits or in the long term, changing names, and organizing the repository, and rewriting markdowns to appear more professional instead of a college level project.